

18 Linear regression and correlation

2

Recall the example:

Note

Example: Cat's Heart Weight

Linear regression of cat's heart weight on body weight, using the dataset *cats* from the MASS library in R.

```
library(MASS)
model1 <- lm(Hwt ~ Bwt, data = cats)
model1
```

Call:

```
lm(formula = Hwt ~ Bwt, data = cats)
```

Coefficients:

(Intercept)	Bwt
-0.3567	4.0341

18.0.1 Estimating the coefficients

The coefficients β_0 and β_1 in the linear model are unknown. The goal is therefore to obtain estimates $\hat{\beta}_0$ and $\hat{\beta}_1$ such that the linear model fits the $n = 144$ data points as closely as possible.

There are several ways to measure *closeness*, however the most common approach involves minimizing the *least squares* criterion.

In least squares, we calculate the difference between the observed value y_i and the predicted value $\hat{y}_i$ obtained from the fitted model: $\hat{y}_i = \hat{\beta}_0 + \hat{\beta}_1 x_i$.

That is, we minimize the mean squared error:

$$\beta_0, \beta_1 = \underset{\beta_1, \beta_0}{\text{arg min}} \text{MSE} = \min \sum_{i=1}^n \frac{(y_i - \hat{y}_i)^2}{n}$$

Let $e_i = y_i - \hat{y}_i$

Handwritten notes:

- $\beta_0 + \beta_1 x_i$ (labeled "line")
- general machine learning terminology
- argument that minimize the obj. function MSE

represents the *i-th residual* for observation *i* then, equivalently our goal is to minimize the residual sum of squares:

Handwritten note: classic linear regression terminology

$$RSS = e_1^2 + e_2^2 + \dots + e_n^2$$

We illustrate these residuals for the *cats* sample in the plot below, shown as vertical lines from each observed data point (in red) to the fitted line.

Call:

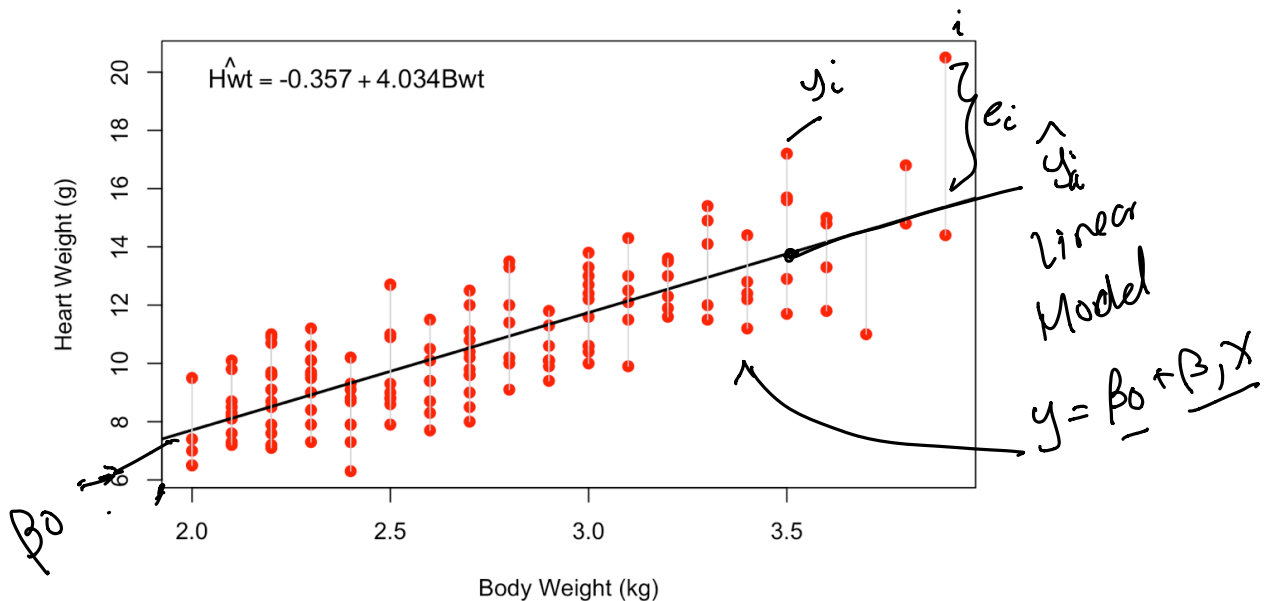
```
lm(formula = Hwt ~ Bwt, data = cats)
```

Coefficients:

(Intercept)	Bwt
-0.3567	4.0341

$$y = \beta_0 + \beta_1 x$$

Simple Linear Regression: Heart Weight vs Body Weight



The *least squares coefficient estimates* $\hat{\beta}_0$ and $\hat{\beta}_1$ for simple linear regression are obtained by minimizing the RSS, which after some calculus yields:

$$\hat{\beta}_1 = \frac{\sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y})}{\sum_{i=1}^n (x_i - \bar{x})^2} = r \left(\frac{s_y}{s_x} \right), \quad (18.1)$$

$$\hat{\beta}_0 = \bar{y} - \hat{\beta}_1 \bar{x},$$

where $\bar{y} \equiv \frac{1}{n} \sum_{i=1}^n y_i$ and $\bar{x} \equiv \frac{1}{n} \sum_{i=1}^n x_i$ are the sample means.

18.0.2 Assessing the accuracy of the coefficient estimates

A natural question about the linear regression model is how accurate the coefficient estimates are — that is, how far $\hat{\beta}_0$ and $\hat{\beta}_1$ are from the true values β_0 and β_1 . To answer this question, we compute the *standard errors* associated with $\hat{\beta}_0$ and $\hat{\beta}_1$ using the following formulas:

← slope

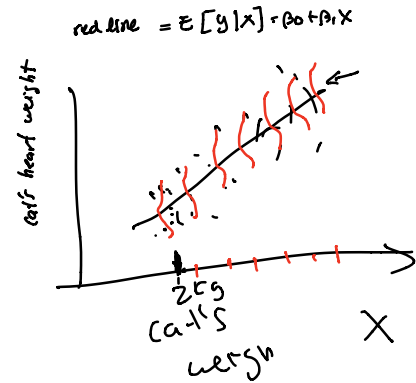
$H_0: \beta_1 = 0$

$H_0: \text{there is no association between } x \text{ and } y$

$$y = \beta_0 + \beta_1 x$$

whole pop. reg. line

$$y_i = \hat{\beta}_0 + \hat{\beta}_1 x_i$$



$$\min \sum (y_i - \hat{y}_i)^2$$

$$\min \sum (y_i - \hat{\beta}_0 - \hat{\beta}_1 x_i)^2$$

$$(1) \quad \frac{\partial MSE}{\partial \hat{\beta}_0} = 2 \sum_i (y_i - \hat{\beta}_0 - \hat{\beta}_1 x_i)^2 (-1) = 0$$

$$(2) \quad \frac{\partial MSE}{\partial \hat{\beta}_1} = 2 \sum_i (y_i - \hat{\beta}_0 - \hat{\beta}_1 x_i) (-x_i) = 0$$

$$(1) \quad \sum y_i - \hat{\beta}_0 \cdot n - \hat{\beta}_1 \sum x_i = 0$$

$$(2) \quad \sum y_i x_i - \hat{\beta}_0 \sum x_i - \hat{\beta}_1 \sum x_i^2 = 0$$

from (1)

$$\bar{y} - \hat{\beta}_0 - \hat{\beta}_1 \bar{x} = 0$$

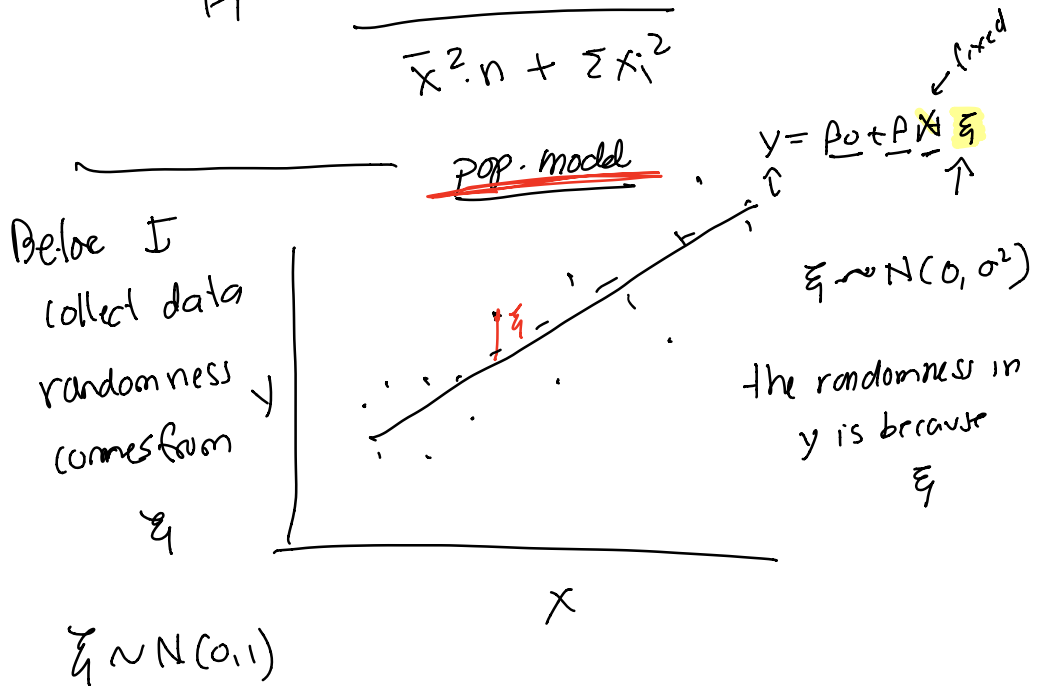
$$\boxed{\hat{\beta}_0 = \hat{\beta}_1 \bar{x} - \bar{y}}$$

plug it in eq (2)

$$\sum x_i y_i - (\hat{\beta}_1 \bar{x} - \bar{y}) \bar{x} \cdot n - \hat{\beta}_1 \sum x_i^2 = 0$$

$$\sum x_i y_i - \hat{\beta}_1 \bar{x}^2 \cdot n + \bar{y} \bar{x} n - \hat{\beta}_1 \sum x_i^2 = 0$$

$$\hat{\beta}_1 = \frac{\sum x_i y_i + \bar{y} \bar{x} \cdot n}{\bar{x}^2 \cdot n + \sum x_i^2}$$



variable

constant

r. variable

$$\hat{\beta}_0 = \hat{\beta}_1 \bar{x} - \bar{y}$$

$$\text{Var}(\hat{\beta}_0) = \bar{x}^2 \text{Var}(\hat{\beta}_1) + \text{Var}(\bar{y}) - 2 \text{Cov}(\hat{\beta}_1 \bar{x}, \bar{y})$$

$$SE(\hat{\beta}_0)^2 = \sigma^2 \left[\frac{1}{n} + \frac{\bar{x}^2}{\sum_{i=1}^n (x_i - \bar{x})^2} \right], \quad SE(\hat{\beta}_1)^2 = \frac{\sigma^2}{\sum_{i=1}^n (x_i - \bar{x})^2} \quad (18.2)$$

where $\sigma^2 = \text{Var}(\epsilon)$ is the variance of the error terms, which is unknown but can be estimated empirically via the RSE defined as follows:

$$\hat{\sigma}^2 = RSE = \sqrt{\frac{RSS}{n-2}}$$

Strictly, these equations are valid only when the errors ϵ_i for each observation have constant variance σ^2 and are uncorrelated, but in practice they provide a good approximation.

Notice that unlike e_i , which are the observed residuals obtained from the fitted model, ϵ_i are the underlying population errors that arise when approximating the relationship between X and Y through a linear model, and are therefore unobserved.

Once we obtain the standard errors of the estimates, we can construct *confidence intervals*. A 95% confidence interval is defined as a range of values such that, with 95% probability, the range will contain the true unknown value of the parameter. Confidence intervals are constructed from the data and have the property that if we take repeated samples and construct a confidence interval for each sample, 95% of those intervals will contain the true unknown value of the parameter. For the linear regression model, the 95% confidence intervals for β_0 and β_1 are approximated as:

$$\left[\hat{\beta}_0 - 2 \cdot SE(\hat{\beta}_0), \hat{\beta}_0 + 2 \cdot SE(\hat{\beta}_0) \right] \quad (18.3)$$

$$\left[\hat{\beta}_1 - 2 \cdot SE(\hat{\beta}_1), \hat{\beta}_1 + 2 \cdot SE(\hat{\beta}_1) \right] \quad (18.4)$$

In the previous example, the standard errors for the intercept and slope were 0.6923 and 0.2503, respectively. The associated confidence intervals are therefore:

```
::: {cell}
```

```
confint(model1)
```

```
::: {cell-output .cell-output-stdout}
```

```

                2.5 %    97.5 %
(Intercept) -1.725163  1.011838
Bwt          3.539343  4.528782
```

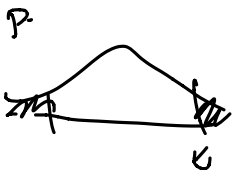
```
::: :::
```

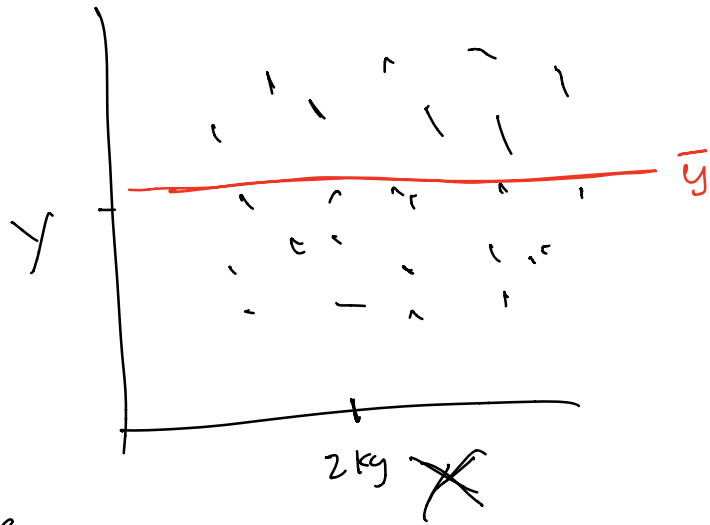
We are thus 95% confident that for every 1 kg increase in body weight, the true average increase in heart weight is between 3.54 and 4.53 grams.

t-test

$$H_0: \beta_1 = 0$$

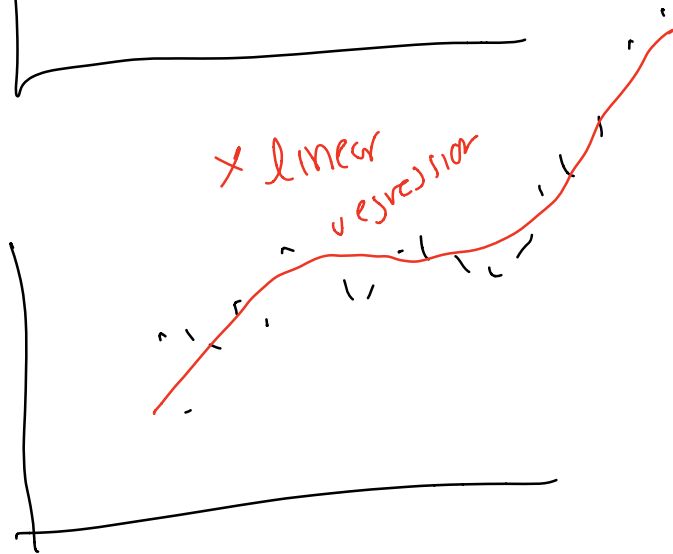
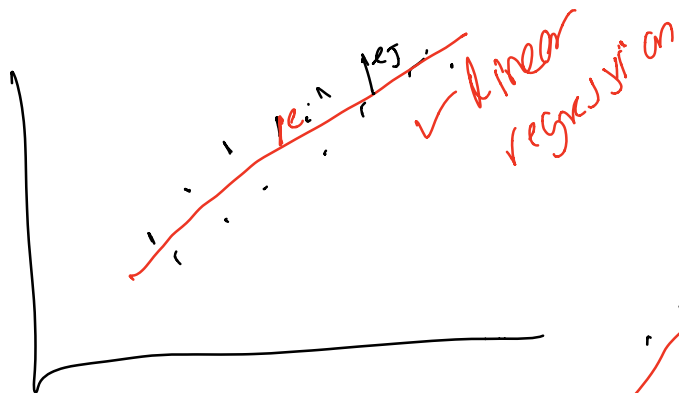
$$t = \frac{\hat{\beta}_1 - 0}{SE(\hat{\beta}_1)}$$





In-the sample

Predictor $\bar{y}$ (you ignore x)



Standard errors are also useful for testing the most common hypothesis in regression — that there is no relationship between X and Y (i.e., that a cat's body weight is not associated with its heart weight). Formally, the *null hypothesis* states that there is no relationship between X and Y , versus the *alternative hypothesis* that there is some relationship between X and Y . Mathematically, this corresponds to testing:

$$\begin{aligned} H_0 : \beta_1 &= 0 \\ \text{versus} \\ H_a : \beta_1 &\neq 0 \end{aligned} \tag{18.5}$$

To test the null hypothesis, we need to determine how far $\hat{\beta}_1$ is from zero to be confident that β_1 is nonzero. This depends on the accuracy of the estimate, given by $SE(\hat{\beta}_1)$. The smaller $SE(\hat{\beta}_1)$ is, the more likely that β_1 differs from zero; conversely, the larger the standard error, the larger $\hat{\beta}_1$ needs to be in order to reject the null hypothesis. In practice, we rely on a *t-statistic*, given by

$$t = \frac{\hat{\beta}_1 - 0}{SE(\hat{\beta}_1)}, \tag{18.6}$$

← uses RSS + has $n-2$ d.f. because we are estimating 2 parameters

which measures how many standard deviations $\hat{\beta}_1$ is from zero. If there is no relationship between X and Y , we expect t to follow a *t*-distribution with $n - 2$ degrees of freedom. The larger the observed value of t relative to the theoretical *t*-distribution, the lower the probability that $\beta_1 = 0$. This probability is called the *p-value*, which in our example was practically zero ($< 2 \times 10^{-16}$), providing strong evidence that $\hat{\beta}_1 \neq 0$, or equivalently that it is *statistically significant*. A small *p-value* indicates that it is unlikely to observe such a substantial association between the predictor and the response by chance alone. We therefore reject the null hypothesis when the *p-value* is sufficiently small — typically below 0.05 or 0.01.



non-constant variance
in ε

heteroskedastic